Notes: Abadie, Angrist, and Imbens (Econometrica, 2002)

Instrumental Variables Estimates of the Effect of Subsidized Training on the Quantiles of Trainee Earnings

Contents

1 Big picture

- **Question.** How can researchers estimate the effect of an endogenous treatment on the *distribution* of outcomes, rather than only on the mean?
- **Application.** The paper studies the Job Training Partnership Act (JTPA), a large U.S. subsidized training programme for economically disadvantaged workers.
- **Key empirical issue.** In the National JTPA Study, offers of training were randomly assigned, but not everyone offered training actually received training. Actual training participation is therefore endogenous, while randomized offer status is a valid instrument.
- **Methodological contribution.** The paper develops an instrumental-variables quantile estimator: the **Quantile Treatment Effects** (QTE) estimator.
- **Causal estimand.** QTE estimates differences in conditional quantiles of potential outcomes for **compliers**: individuals whose treatment status is changed by the instrument.
- **Why quantiles matter.** Average treatment effects can hide whether the programme helps low earners, middle earners, or high earners. Distributional effects matter for welfare, inequality, targeting, and program design.
- **Main econometric idea.** Under IV assumptions, compliers can be recovered in expectation by a weighting function. Replacing the unobserved complier indicator with an estimable conditional probability yields a weighted quantile regression problem.
- **Computational contribution.** A naive IV quantile objective can be nonconvex because the IV weights may be negative. The paper shows how to transform it into a convex weighted quantile regression by conditioning on observed variables.
- **Main empirical finding for women.** JTPA training raises earnings quantiles for women throughout the distribution, with the largest proportional effects at low quantiles.
- **Main empirical finding for men.** JTPA training has little effect on the lower quantiles of male trainee earnings, but has positive effects in the upper half of the distribution.
- **Economic takeaway.** Mean IV estimates are incomplete. For the JTPA, training appears to help women broadly and low-earning women especially, but male effects are concentrated higher in the earnings distribution.

2 Data, treatment, and causal framework

2.1 Observed variables

The data consist of independent observations on

$$(Y_i, D_i, Z_i, X_i), \qquad i = 1, \ldots, n, \tag{1}$$

where:

- Y_i is the scalar outcome; in the application it is 30-month earnings.
- $D_i \in \{0, 1\}$ is the treatment indicator; in the application it indicates actual JTPA training participation.
- $Z_i \in \{0, 1\}$ is the instrument; in the application it indicates randomized offer of JTPA training.
- X_i is a vector of baseline covariates, including demographics and pre-randomization characteristics.

Interpretation: The paper separates *being offered* training from *taking* training. Random assignment makes Z exogenous; actual participation D is endogenous because individuals decide whether to participate.

2.2 Potential outcomes and potential treatment status

Potential outcomes are indexed by treatment status:

$$Y_1 = \text{earnings if trained}, \qquad Y_0 = \text{earnings if not trained}. \tag{2}$$

Potential treatment choices are indexed by assignment status:

$$D_1 = \text{treatment status if offered training}, \qquad D_0 = \text{treatment status if not offered training}. \tag{3}$$

The observed treatment and outcome are

$$D = ZD_1 + (1 - Z)D_0, \qquad Y = DY_1 + (1 - D)Y_0. \tag{4}$$

Interpretation: The causal object is not just one average effect. The paper asks how training shifts the quantiles of Y_1 relative to Y_0 for the relevant subpopulation.

2.3 IV assumptions

The paper assumes that for almost all values of X:

1. **Independence:**
$$(Y_1, Y_0, D_1, D_0) \perp\!\!\!\perp Z \mid X. \tag{5}$$

2. **Nontrivial assignment:**
$$P(Z = 1 \mid X) \in (0, 1). \tag{6}$$

3. **First stage:**
$$E[D_1 \mid X] \neq E[D_0 \mid X]. \tag{7}$$

4. **Monotonicity:**
$$P(D_1 \geq D_0 \mid X) = 1. \tag{8}$$

Interpretation: Independence and exclusion are plausible because training offers were randomized. Monotonicity means the offer does not make anyone less likely to receive training. In the JTPA context, it is especially plausible because almost nobody in the control group received JTPA training.

2.4 Compliers

The complier group is

$$D_1 > D_0. \tag{9}$$

These are individuals whose training status is changed by the randomized offer.

The key implication is:

$$(Y_1, Y_0) \perp\!\!\!\perp D \mid X,\ D_1 > D_0. \tag{10}$$

Interpretation: Within the complier population, treatment status is as-good-as random conditional on X. The difficulty is that we do not observe whether a given individual is a complier.

2.5 Why one-sided noncompliance matters in JTPA

In many social experiments, including approximately in JTPA,

$$Z = 0 \Rightarrow D = 0. \tag{11}$$

This means that individuals in the control group are essentially excluded from the treatment.

Interpretation: With one-sided noncompliance, the complier population is close to the treated population. Therefore, QTE estimates can be interpreted as distributional effects of training for trainees.

3 Quantile treatment effects model

3.1 Conditional quantile model

For a quantile index $\theta \in (0, 1)$, the paper assumes a linear conditional quantile model for compliers:

$$Q_\theta(Y \mid X, D, D_1 > D_0) = \alpha_\theta D + X'\beta_\theta. \tag{12}$$

- $Q_\theta(\cdot)$ denotes the conditional θ-quantile.
- α_θ is the treatment effect on the θ-quantile.
- β_θ controls for baseline covariates.

Interpretation: The coefficient α_θ is the difference between conditional quantiles of trained and untrained potential outcomes for compliers. It is not the θ-quantile of the individual-level treatment effect $Y_1 - Y_0$.

3.2 Quantile regression check function

Define the check function

$$\rho_\theta(u) = (\theta - \mathbf{1}\{u < 0\})\, u. \tag{13}$$

If the complier group were observed, the target parameter would solve

$$(\alpha_\theta, \beta_\theta) = \arg\min_{\alpha,\beta} E\left[\rho_\theta\left(Y - \alpha D - X'\beta\right) \mid D_1 > D_0\right]. \tag{14}$$

Interpretation: This is ordinary quantile regression inside the complier subpopulation. The main problem is that the complier subpopulation is latent.

3.3 Recovering compliers by IV weights

Let

$$\pi_0(X) = P(Z = 1 \mid X). \tag{15}$$

The paper defines

$$K(D, Z, X) = 1 - \frac{D(1 - Z)}{1 - \pi_0(X)} - \frac{(1 - D)Z}{\pi_0(X)}. \tag{16}$$

This weight has the property that, for suitable functions $h(Y, D, X)$,

$$E[h(Y, D, X) \mid D_1 > D_0] = \frac{E[K(D, Z, X)h(Y, D, X)]}{P(D_1 > D_0)}. \tag{17}$$

Interpretation: The weight K recovers complier moments in expectation. However, K can be negative, which creates a computational problem for quantile regression.

3.4 Population QTE objective

Using the IV weight, the population objective becomes

$$(\alpha_\theta, \beta_\theta) = \arg\min_{\alpha,\beta} E\left[K(D, Z, X)\rho_\theta(Y - \alpha D - X'\beta)\right]. \tag{18}$$

Interpretation: This objective identifies the complier quantile treatment effect, but its sample analogue is not globally convex because K can be negative.

3.5 Convex transformation using conditional expectation

Let

$$U = (Y, D, X), \qquad v_0(U) = P(Z = 1 \mid U). \tag{19}$$

The paper replaces K with

$$K_v(U) = E[K(D, Z, X) \mid U] = 1 - \frac{D(1 - v_0(U))}{1 - \pi_0(X)} - \frac{(1 - D)v_0(U)}{\pi_0(X)}. \tag{20}$$

Under the IV assumptions,

$$K_v(U) = P(D_1 > D_0 \mid U) \geq 0. \tag{21}$$

Interpretation: This is the key computational step. Conditional expectation converts the possibly negative IV weight into a nonnegative weight equal to the conditional probability of being a complier.

3.6 Sample QTE estimator

Let

$$W = (D, X')'. \tag{22}$$

The feasible QTE estimator solves

$$\hat{\delta}_\theta = \arg\min_\delta \frac{1}{n} \sum_{i=1}^{n} \hat{K}_v(U_i)\rho_\theta(Y_i - W_i'\delta), \tag{23}$$

where

$$\delta_\theta = (\alpha_\theta, \beta_\theta')'. \tag{24}$$

Interpretation: Once $\hat{K}_v$ is estimated, this is a weighted quantile regression with nonnegative weights. It can be solved as a standard linear programming problem.

4 First-step estimation and inference

4.1 Estimating the nuisance functions

The estimator requires two nuisance functions:

$$\pi_0(X) = P(Z = 1 \mid X), \qquad v_0(U) = P(Z = 1 \mid Y, D, X). \tag{25}$$

In the paper's asymptotic theory:

- X is discrete or approximated by discrete cells.
- $\pi_0(X)$ is estimated by cell averages.
- $v_0(U)$ is estimated nonparametrically within (X, D) cells using a power-series approximation in Y.

Interpretation: Estimating v_0 is what makes the procedure more involved than ordinary quantile regression or 2SLS.

4.2 Asymptotic distribution

Let

$$\varepsilon_\theta = Y - W'\delta_\theta. \tag{26}$$

Under regularity conditions,

$$\sqrt{n}(\hat{\delta}_\theta - \delta_\theta) \xrightarrow{d} N(0, Q_\theta). \tag{27}$$

The variance matrix has the sandwich form

$$Q_\theta = J_\theta^{-1}\Omega_\theta J_\theta^{-1}. \tag{28}$$

Interpretation: The variance must account for both second-stage weighted quantile regression noise and first-stage estimation of the complier weights.

4.3 Kernel-based variance estimation

The paper estimates J_θ using a kernel approximation to the conditional density of the residual at zero:

$$\hat{J}_\theta = \frac{1}{n}\sum_{i=1}^{n} \hat{K}_v(U_i)\frac{1}{h}\varphi\left(\frac{Y_i - W_i'\hat{\delta}_\theta}{h}\right) W_i W_i'. \tag{29}$$

Interpretation: This is analogous to standard quantile regression inference: the density of the residual at the quantile affects the curvature of the objective function.

4.4 What the theory adds

- It gives a causal interpretation to quantile effects under IV assumptions.
- It converts a nonconvex IV quantile problem into a convex weighted quantile regression.
- It provides asymptotic distribution theory allowing the first step to be nonparametric.
- It nests ordinary quantile regression when treatment is exogenous.

Interpretation: The QTE estimator is the distributional analogue of IV/2SLS for a binary endogenous treatment.

5 Application to JTPA training

5.1 Program background

- The Job Training Partnership Act began funding training in October 1983.
- Title II of JTPA served economically disadvantaged individuals.
- At the time of the National JTPA Study, Title II served roughly 1 million participants per year.
- The annual program cost was about $1.6 billion.
- The National JTPA Study randomized eligible applicants into a treatment offer group and a control group.

Interpretation: Random assignment solves the problem of selection into being offered training. IV is still needed because not everyone offered training actually takes it.

5.2 Sample and covariates

The application uses adult men and adult women separately. The main outcome is

$$Y = \text{earnings during the 30 months after random assignment.} \tag{30}$$

Key covariates include:

- high school or GED status;
- race and Hispanic ethnicity;
- marital status;
- weak prior labor force attachment;
- AFDC receipt for women;
- service strategy indicators.

Interpretation: Separate estimates for men and women are important because the JTPA literature and the paper's results show markedly different distributional impacts by gender.

5.3 Comparison estimators

The paper compares four types of estimates:

1. simple assignment-status contrasts;
2. OLS estimates of training effects;
3. 2SLS estimates using assignment as an instrument for training;
4. QTE estimates using the new IV quantile procedure.

Interpretation: OLS and ordinary quantile regression treat training as exogenous. 2SLS and QTE use the randomized offer as an instrument.

6 Tables and figures

6.1 Table I: Means and standard deviations

Read:

- For men, the full sample contains 5,102 observations; for women, it contains 6,102 observations.
- Around 62% of treatment-assigned men and 66% of treatment-assigned women received training.
- Treatment status raises 30-month earnings by simple trainee versus non-trainee comparisons: about $\$3,970$ for men and $\$2,133$ for women.
- Assignment-status differences are smaller than treatment-status differences because not all assigned to treatment participate.
- Baseline characteristics are broadly balanced by randomized assignment but differ more by actual training status.

Economic message: Random assignment balances covariates, but actual training participation is selective. This motivates using assignment as an instrument for training.

TABLE I
MEANS AND STANDARD DEVIATIONS

	Entire Sample	Assignment: Treatment	Assignment: Control	Assignment: Diff. (t-stat.)	Treatment: Trainees	Treatment: Non-trainees	Treatment: Diff. (t-stat.)
A. Men							
Number of observations	5,102	3,399	1,703		2,136	2,966	
Treatment							
Training	.42	.62	.01	.61			
	[.49]	[.48]	[.11]	(70.34)			
Outcome variable							
30 month earnings	19,147	19,520	18,404	1,116	21,455	17,485	3,970
	[19,540]	[19,912]	[18,760]	(1.96)	[19,864]	[19,135]	(7.15)
Baseline Characteristics							
Age	32.91	32.85	33.04	−.19	32.76	33.02	−.26
	[9.46]	[9.46]	[9.45]	(−.67)	[9.64]	[9.32]	(−.95)
High school or GED	.69	.69	.69	−.00	.71	.68	.03
	[.45]	[.45]	[.45]	(−.12)	[.44]	[.45]	(2.46)
Married	.35	.36	.34	.02	.37	.34	.03
	[.47]	[.47]	[.46]	(1.64)	[.47]	[.46]	(2.82)
Black	.25	.25	.25	.00	.26	.25	.01
	[.44]	[.44]	[.44]	(.04)	[.44]	[.43]	(.48)
Hispanic	.10	.10	.09	.01	.10	.09	.01
	[.30]	[.30]	[.29]	(.70)	[.31]	[.29]	(1.60)
Worked less than	.40	.40	.40	.00	.40	.40	−.00

Table I, Panel A: men

TABLE I—*Continued*

	Entire Sample	Assignment: Treatment	Assignment: Control	Assignment: Diff. (t-stat.)	Treatment: Trainees	Treatment: Non-Trainees	Treatment: Diff. (t-stat.)
B. Women							
Number of observations	6,102	4,088	2,014		2,722	3,380	
Treatment							
Training	.45	.66	.02	.64			
	[.50]	[.47]	[.13]	(80.24)			
Outcome Variable							
30 month earnings	13,029	13,439	12,197	1,242	14,211	12,078	2,133
	[13,415]	[13,614]	[12,964]	(3.46)	[13,550]	[13,230]	(6.18)
Baseline Characteristics							
Age	33.33	33.33	33.35	−.02	33.11	33.52	−.41
	[9.78]	[9.77]	[9.81]	(−.09)	[9.71]	[9.84]	(−1.62)
High school or GED	.72	.73	.70	.03	.75	.70	.05
	[.43]	[.43]	[.44]	(2.01)	[.42]	[.45]	(5.07)
Married	.22	.22	.21	.01	.22	.21	.01
	[.40]	[.40]	[.39]	(1.55)	[.40]	[.39]	(1.35)
Black	.26	.27	.26	.01	.26	.27	−.01
	[.44]	[.44]	[.44]	(.95)	[.44]	[.44]	(−.97)
Hispanic	.12	.12	.12	−.00	.12	.11	.01
	[.32]	[.32]	[.33]	(−.89)	[.33]	[.32]	(1.29)
Worked less than 13 weeks in past year	.52	.52	.52	−.00	.51	.53	−.02
	[.47]	[.47]	[.47]	(−.08)	[.47]	[.47]	(−1.52)
AFDC	.31	.30	.31	−.01	.32	.30	.02
	[.46]	[.46]	[.46]	(−1.03)	[.47]	[.46]	(1.92)

Table I, Panel B: women

6.2 Table II: Quantile regression and OLS estimates

Read:

- Table II treats training as exogenous and reports OLS and quantile regression estimates.
- For men, OLS estimates a training effect of \$3,754, but quantile effects range from \$1,187 at the 0.15 quantile to \$4,806 at the 0.85 quantile.
- For women, OLS estimates a training effect of \$2,215, and quantile effects are positive throughout the distribution.
- Proportional effects are especially large at low quantiles because baseline earnings are low there.

Economic message: Ordinary quantile regression suggests distributional heterogeneity, but it does not correct for endogenous participation in training.

6.3 Table III: QTE and 2SLS estimates

Read:

- Table III uses randomized assignment as an instrument for training.
- For men, the 2SLS mean effect is \$1,593, smaller than the OLS effect.
- Male QTE estimates are small at the lower quantiles: \$121 at 0.15 and \$702 at 0.25.
- Male QTE estimates become larger above the median: \$3,131 at 0.75 and \$3,378 at 0.85.
- For women, the 2SLS mean effect is \$1,780.
- Female QTE estimates are positive at all reported quantiles, with larger proportional impacts at low quantiles.

Economic message: The IV distributional results differ from ordinary quantile regression. For men, training appears to affect upper-tail earnings more than lower-tail earnings. For women, effects are broad and proportionally strongest near the bottom.

TABLE II

Quantile Regression and OLS Estimates

Dependent Variable: 30-month Earnings

		Quantile				
	OLS	0.15	0.25	0.50	0.75	0.85
A. Men						
Training	3,754	1,187	2,510	4,420	4,678	4,806
	(536)	(205)	(356)	(651)	(937)	(1,055)
% Impact of Training	21.2	135.6	75.2	34.5	17.2	13.4
High school or GED	4,015	339	1,280	3,665	6,045	6,224
	(571)	(186)	(305)	(618)	(1,029)	(1,170)
Black	−2,354	−134	−500	−2,084	−3,576	−3,609
	(626)	(194)	(324)	(684)	(1,087)	(1,331)
Hispanic	251	91	278	925	−877	−85
	(883)	(315)	(512)	(1,066)	(1,769)	(2,047)
Married	6,546	587	1,964	7,113	10,073	11,062
	(629)	(222)	(427)	(839)	(1,046)	(1,093)
Worked less than 13	−6,582	−1,090	−3,097	−7,610	−9,834	−9,951
weeks in past year	(566)	(190)	(339)	(665)	(1,000)	(1,099)
Constant	9,811	−216	365	6,110	14,874	21,527
	(1,541)	(468)	(765)	(1,403)	(2,134)	(3,896)
B. Women						
Training	2,215	367	1,013	2,707	2,729	2,058
	(334)	(105)	(170)	(425)	(578)	(657)
% Impact of Training	18.5	60.8	44.4	32.3	14.5	8.09
High school or GED	3,442	166	681	2,514	5,778	6,373
	(341)	(99)	(156)	(396)	(606)	(762)
Black	−544	22	−60	−129	−866	−1,446
	(397)	(115)	(188)	(451)	(679)	(869)
Hispanic	−1,151	−31	−222	−995	−1,620	−1,503
	(488)	(130)	(194)	(546)	(911)	(992)
Married	−667	−213	−392	−758	−1,048	−902
	(436)	(127)	(209)	(522)	(785)	(970)
Worked less than 13	−5,313	−1,050	−3,240	−6,872	−7,670	−6,470
weeks in past year	(370)	(137)	(289)	(522)	(672)	(787)
AFDC	−3,009	−398	−1,047	−3,389	−4,334	−3,875
	(378)	(107)	(174)	(468)	(737)	(834)
Constant	10,361	649	2,633	8,417	16,498	20,689
	(815)	(255)	(490)	(966)	(1,554)	(1,232)

Note: The table reports OLS and quantile regression estimates of the effect of training on earnings. The specification also includes indicators for service strategy recommended, age group, and second follow-up survey. Robust standard errors are reported in parentheses.

TABLE III

Quantile Treatment Effects and 2SLS Estimates

Dependent Variable: 30-month Earnings

		Quantile				
	2SLS	0.15	0.25	0.50	0.75	0.85
A. Men						
Training	1,593	121	702	1,544	3,131	3,378
	(895)	(475)	(670)	(1,073)	(1,376)	(1,811)
% Impact of Training	8.55	5.19	12.0	9.64	10.7	9.02
High school or GED	4,075	714	1,752	4,024	5,392	5,954
	(573)	(429)	(644)	(940)	(1,441)	(1,783)
Black	−2,349	−171	−377	−2,656	−4,182	−3,523
	(625)	(439)	(626)	(1,136)	(1,587)	(1,867)
Hispanic	335	328	1,476	1,499	379	1,023
	(888)	(757)	(1,128)	(1,390)	(2,294)	(2,427)
Married	6,647	1,564	3,190	7,683	9,509	10,185
	(627)	(596)	(865)	(1,202)	(1,430)	(1,525)
Worked less than 13	−6,575	−1,932	−4,195	−7,009	−9,289	−9,078
weeks in past year	(567)	(442)	(664)	(1,040)	(1,420)	(1,596)
Constant	10,641	−134	1,049	7,689	14,901	22,412
	(1,569)	(1,116)	(1,655)	(2,361)	(3,292)	(7,655)
B. Women						
Training	1,780	324	680	1,742	1,984	1,900
	(532)	(175)	(282)	(645)	(945)	(997)
% Impact of Training	14.6	35.5	23.1	18.4	10.1	7.39
High school or GED	3,470	262	768	2,955	5,518	5,905
	(342)	(178)	(274)	(643)	(930)	(1026)
Black	−554	0	−123	−401	−1,423	−2,119
	(397)	(204)	(318)	(724)	(949)	(1,196)
Hispanic	−1,145	−73	−138	−1,256	−1,762	−1,707
	(488)	(217)	(315)	(854)	(1,188)	(1,172)
Married	−652	−233	−532	−796	38	−109
	(437)	(221)	(352)	(846)	(1,069)	(1,147)
Worked less than 13	−5,329	−1,320	−3,516	−6,524	−6,608	−5,698
weeks in past year	(370)	(254)	(430)	(781)	(931)	(969)
AFDC	−2,997	−406	−1,240	−3,298	−3,790	−2,888
	(378)	(189)	(301)	(743)	(1,014)	(1,083)
Constant	10,538	984	3,541	9,928	15,345	20,520
	(828)	(547)	(837)	(1,696)	(2,387)	(1,687)

Note: The table reports 2SLS and QTE estimates of the effect of training on earnings. Assignment status is used as an instrument for training. The specification also includes indicators for service strategy recommended, age group, and second follow-up survey. Robust standard errors are reported in parentheses.

6.4 Table IV: Earnings quantiles without training

Read:

- Table IV reports quantiles of the distribution of earnings without training, Y_0, for trainees and nontrainees.
- For men, trainee Y_0 quantiles are higher than nontrainee Y_0 quantiles, especially at low quantiles.
- Conditional on covariates, male trainees still appear positively selected relative to nontrainees at the bottom of the distribution.
- For women, trainee and nontrainee Y_0 quantiles are more similar.

Economic message: Selection into training differs by gender. Low-potential-earnings men appear less likely to receive JTPA training, which helps explain why male QTE effects are small at low quantiles.

TABLE IV

30-month Earnings Quantiles without Training

	Quantile				
	0.15	0.25	0.50	0.75	0.85
A. Men					
Unconditional					
Trainees	1,368	4,797	15,362	29,947	39,171
Non-Trainees	142	1,623	11,449	27,855	37,524
Conditional					
Trainees	2,329	5,859	16,016	29,294	37,448
Non-Trainees	876	3,337	12,814	27,137	35,778
B. Women					
Unconditional					
Trainees	120	1,380	8,722	20,392	26,538
Non-Trainees	0	840	7,566	20,241	26,630
Conditional					
Trainees	906	2,924	9,482	19,719	25,719
Non-Trainees	605	2,281	8,394	18,856	25,444

Note: The table reports quantiles of the distribution of earnings without training (Y_0) for trainees and nontrainees. Since there was almost perfect compliance in the control group, quantiles for trainees are given by quantiles for compliers. The rows labeled *Unconditional* report unconditional quantiles. To produce the results for the rows labeled *Conditional*, the conditional quantile estimates of Table III were evaluated at the mean of the covariates for the treated with the *Training* indicator set to zero.

7 Short summary

- The paper develops an IV method for estimating treatment effects on quantiles.
- The method applies to binary endogenous treatment with a binary instrument.
- The causal estimand is a quantile treatment effect for compliers.
- The estimator uses weights that recover compliers in expectation.
- A naive IV quantile objective is nonconvex because IV weights can be negative.
- Taking conditional expectations produces nonnegative complier-probability weights.
- The feasible estimator is a weighted quantile regression with first-step nonparametric estimation of nuisance functions.
- The application studies JTPA training using random assignment as an instrument for actual training.
- For women, training raises earnings throughout the distribution, with largest proportional effects at low quantiles.
- For men, training effects are concentrated in the upper half of the earnings distribution.
- The results show why mean effects alone can be misleading for programme evaluation.

8 Conclusion

8.1 Core conclusion

Abadie, Angrist, and Imbens provide a causal IV framework for estimating treatment effects on quantiles. Their estimator extends the logic of local average treatment effects to distributional outcomes. In the JTPA application, the distributional estimates reveal patterns that mean IV estimates cannot show.

8.2 Economic intuition

Random assignment of training offers identifies variation in actual training participation. Among compliers, treatment is as-good-as randomly assigned. The QTE estimator uses this insight to estimate how treatment shifts different parts of the earnings distribution. The method shows whether a programme helps low earners, high earners, or both.

8.3 Takeaway

The paper is important because it turns IV from a mean-effects tool into a distributional evaluation tool. For JTPA, this matters substantively: training appears to help women across the distribution but affects men's earnings mainly in the upper half. The welfare and targeting implications are therefore different from what one would infer from average effects alone.